Software Engineering

Software Quality
(Product vs Process)

Overview

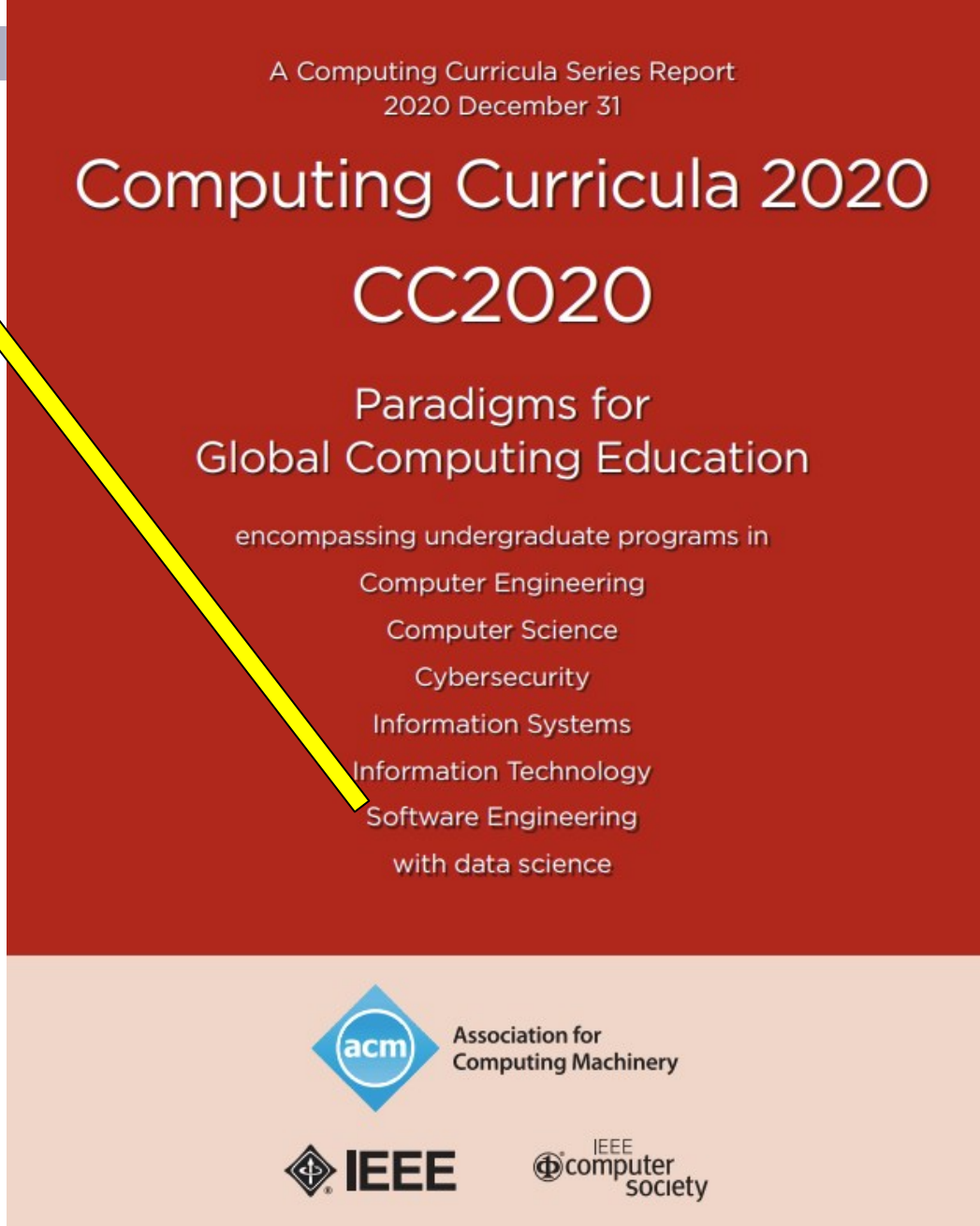
- Motivation
- Software process quality
- Software product quality

Soft. Eng. Competences

- Software Requirements
- Software Design
- Software Construction
- Software Testing
- Software Sustainment
- Software Process and Life Cycle
- Software Systems Engineering
- Software Quality
- Software Security
- Software Safety
- Software Configuration Manag.
- Software Measurement
- Human-Computer Interaction
- Project Management
- Behavioral Attributes

CDSI

IPM



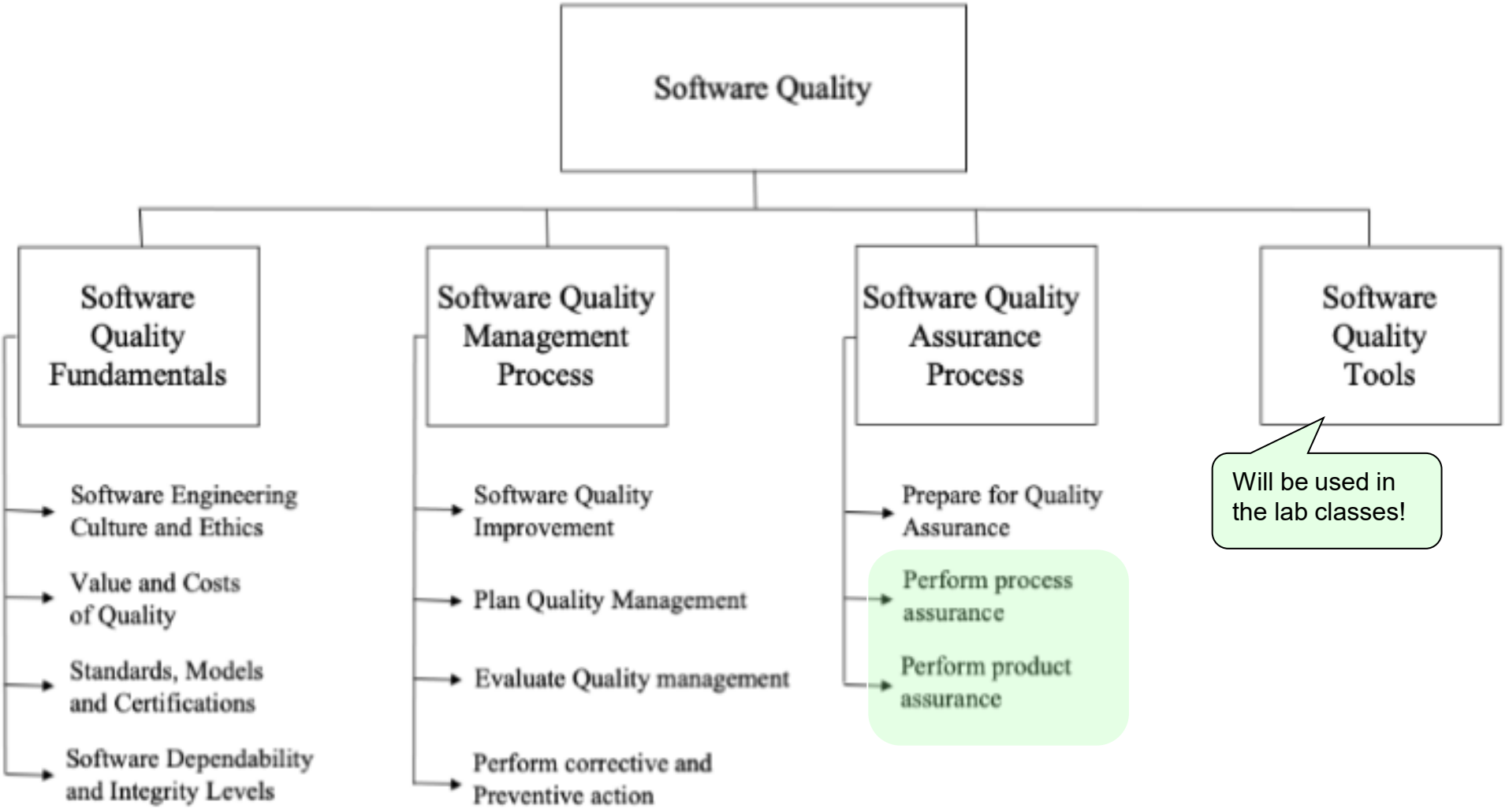




1. Software Requirements
2. Software Architecture
3. Software Design
4. Software Construction
5. Software Testing
6. Software Engineering Operations
7. Software Maintenance
8. Software Configuration Management
9. Software Engineering Management
10. Software Engineering Models and Methods
11. Software Engineering Process
12. Software Quality
13. Software Security
14. Software Engineering Economics
15. Software Engineering Professional
16. Computing Foundations
17. Mathematical Foundations
18. Engineering Foundations

Software quality may refer to the desirable characteristics of software products, to the extent to which a particular software product has those characteristics (**software product quality**), and to the processes, tools and techniques used to achieve those characteristics (**software process quality**) [SWEBOK 4.0]

SWEBOK®


Software Quality
Software Quality Fundamentals
Software Quality Management Process
Software Quality Assurance Process
Software Quality Tools
Software Engineering Culture and Ethics
Value and Costs of Quality
Standards, Models and Certifications
Software Dependability and Integrity Levels
Software Quality Improvement
Plan Quality Management
Evaluate Quality management
Perform corrective and Preventive action
Prepare for Quality Assurance
Perform process assurance
Perform product assurance
Will be used in the lab classes!

Software Quality initiatives





International Conference on the Quality of Information and Communications Technology

Standard Series:
ISO/IEC 25000 (SQuaRE)
System and Software Quality Requirements and Evaluation



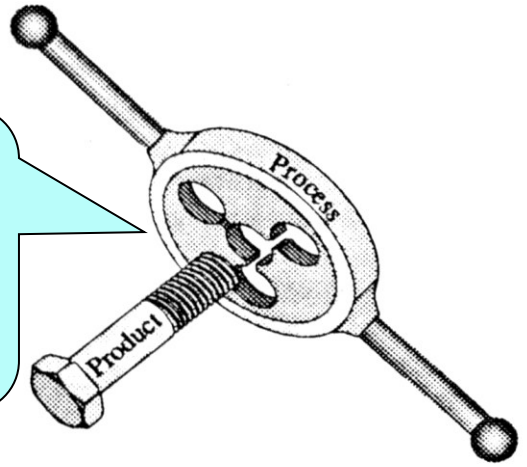




Software process quality

Process Quality

In this visual metaphor, the rotating tool (representing the process) is carving (i.e. conditioning) the pitch quality of the screw thread (representing the product)



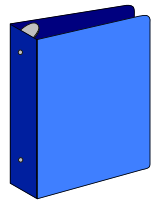
- The quality of the software process is a necessary, but not sufficient, condition for product quality!
  - The opposite (i.e., achieving product quality with a bad/ad-hoc software process) only happens by chance, as when the development team includes very talented members, comparable to “artists” in a craft context

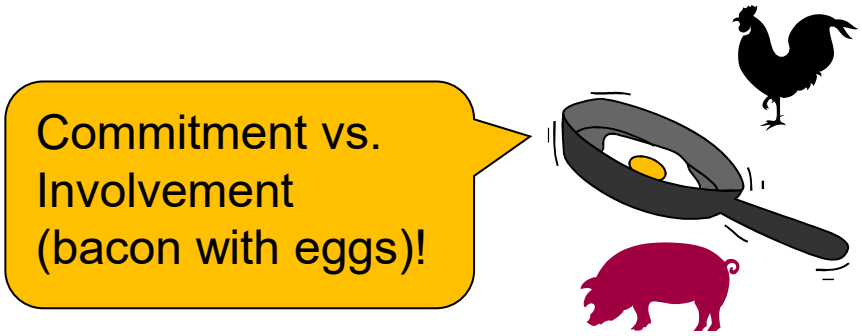
Process Quality is achieved by the implementation of an appropriate **Quality Management System (QMS)**

Quality Management System (QMS)

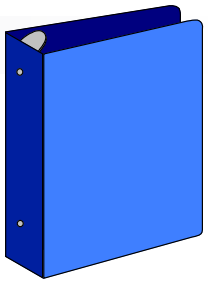
- Is a set of processes, organization, resources and procedures used to produce products of consistent quality which meet defined specifications

- It must:
  - be perfectly defined and documented - **Quality Manual**.
  - clearly identify the management commitment
  - include a definition of strategies to develop
  - detail the steps to fulfill





What must be in a Quality Manual?



- Any Software Quality Manual must include, at least, the following themes:
  - Definition of development life-cycle and documents to produce in each phase
  - Organizational structure of projects
  - Tasks distribution (responsibility assignment)
  - Execution schedules and resource management
  - Configuration management
  - Risks identification and contingency plans
  - Mechanisms of verification and validation (V&V)

Certification of a QMS

- To demonstrate Quality, organizations must submit their QMS to third-party certification (i.e., to a certifying body)
- Quality system registration encompasses a periodic assessment of the QMS
  - Major audit on a triennial basis
  - Minor audit on a yearly basis
- Auditors use a reference standard to report non-compliances
  - Standards used: ISO/IEC 9001, ISO/IEC 33000 or CMMI
  - Auditors check for QMS controllable elements (i.e. process evidences)

QMS controllable elements

- Personnel
- Material
- Means
- Documentation

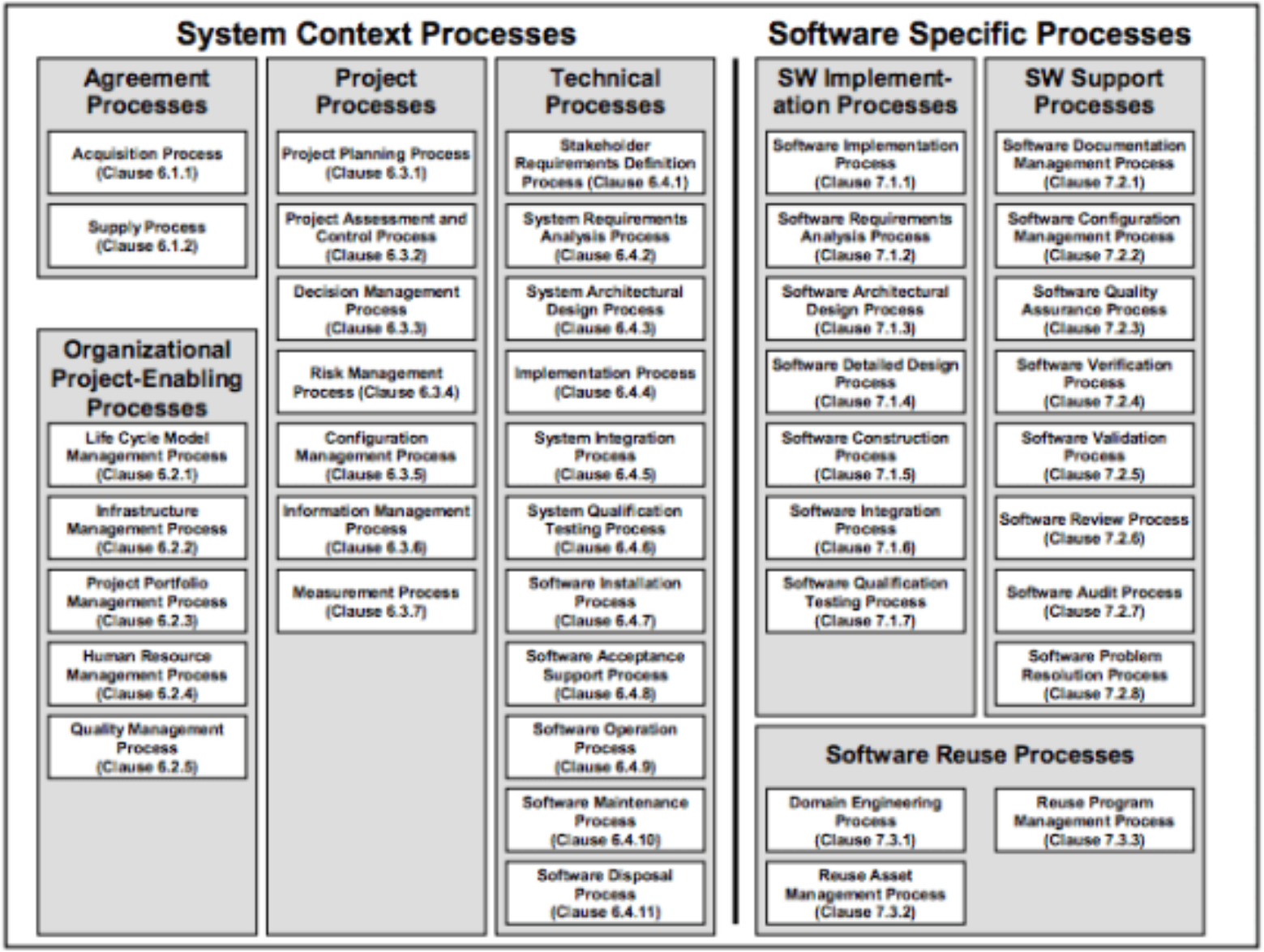
to perform or supervise the production tasks [Competence, training, identification, motivation, attributes]

on which to apply production processes (component, sub-assembly, etc.) [Type, condition, capability, quantity, identification]

tools, equipment and facilities required to perform the task [Type, condition, capability, identification, location, environment]

input/output (specifications, drawings, records, reports, orders, etc.) [Content, control, condition, identification, distribution]

ISO/IEC 12207 Software life cycle processes



ISO/IEC 33000 Process Quality Models

- ISO/IEC 33000 family defines the framework for process maturity assessment, and improvement
- Its generic core parts are:
  - ISO/IEC 33001: Concepts and terminology
  - ISO/IEC 33002: Requirements for assessments
  - ISO/IEC 33020: Process measurement framework
- Those parts can be applied to *any process domain*:
  - software, systems, IT service management, cybersecurity, etc.
- This part is software specific:
  - ISO/IEC 33061 — *Process Assessment Model for Software Life Cycle Processes based on ISO/IEC 12207*

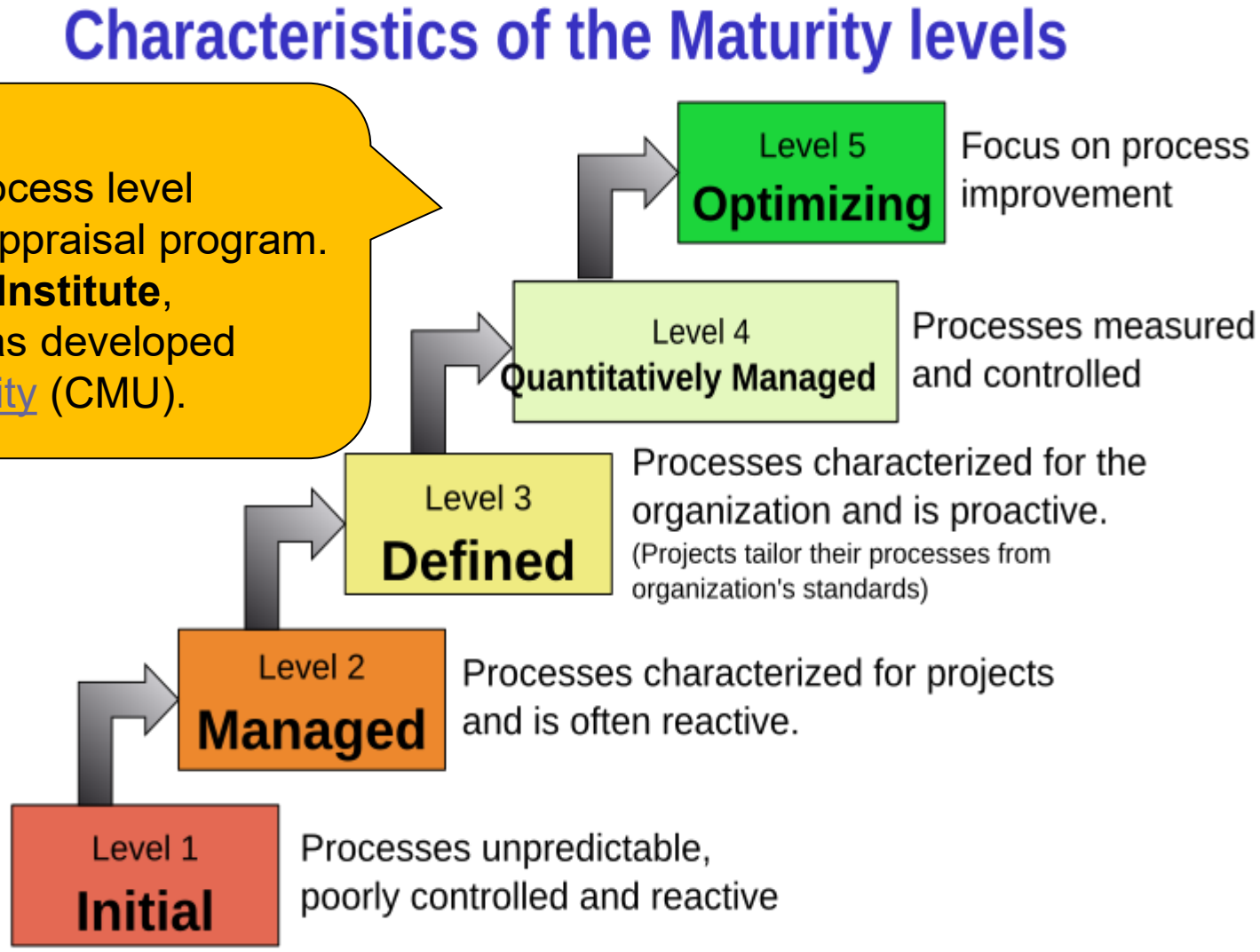
Process Capability Levels in ISO/IEC 33020

- ISO/IEC 33020 defines six process capability levels:
  - Level 0 – Incomplete: Process not implemented or fails to achieve purpose
  - Level 1 – Performed: Process achieves defined outcomes
  - Level 2 – Managed: Planned, monitored, and controlled
  - Level 3 – Established: Standardized and institutionalized
  - Level 4 – Predictable: Quantitatively measured and controlled
  - Level 5 – Innovating/Optimizing: Continuously improved and adapted

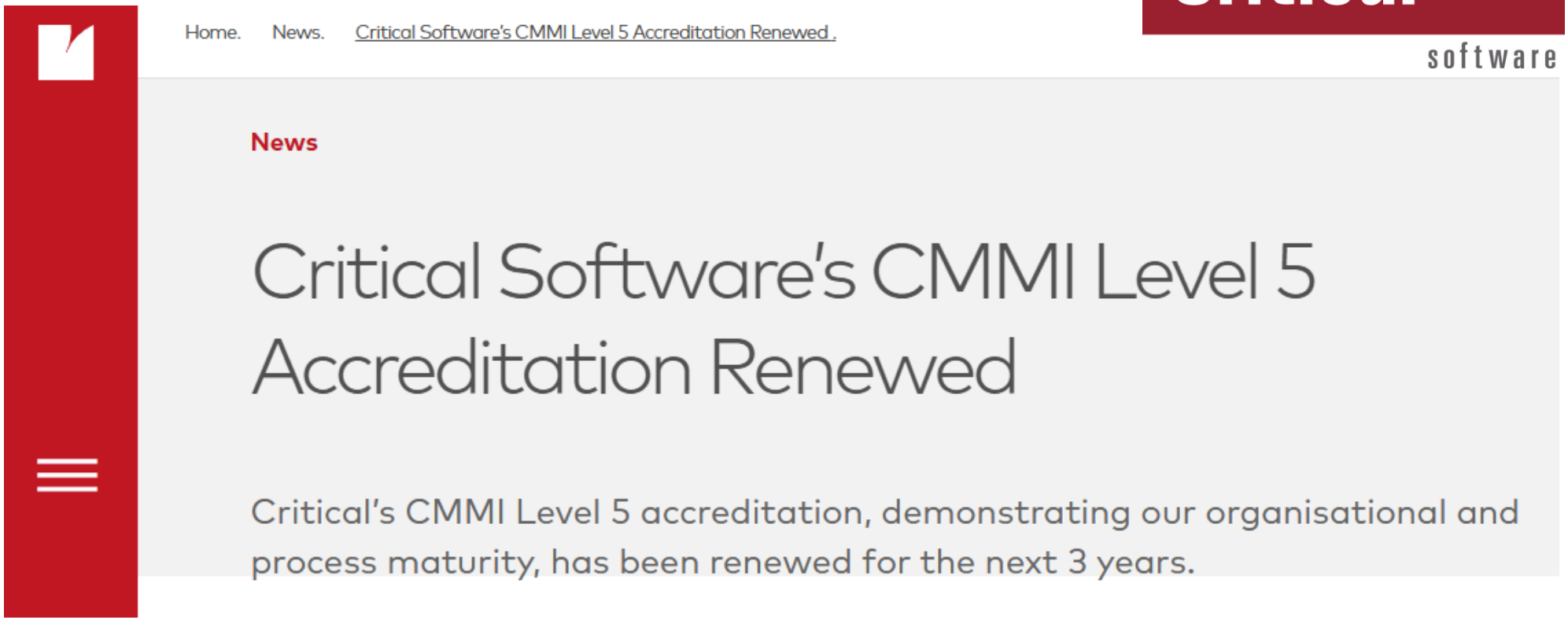
These capability levels were inspired on CMMI (see next slide)

CMMI Process Quality Framework

Capability Maturity Model Integration (**CMMI**) is a process level improvement training and appraisal program. Administered by the **CMMI Institute**, a subsidiary of ISACA, it was developed at Carnegie Mellon University (CMU).



Example from Portugal



Accredited certification bodies (in Portugal)

CERTIFICATION BODY
APCER - Associação Portuguesa de Certificação
SGS ICS - Serviços Internacionais de Certificação, Lda.
Bureau Veritas Certification Portugal, Unipessoal, Lda.
EIC - Empresa Internacional de Certificação, S.A.
Tuv-Rheinland Portugal, Inspecções Técnicas, Unipessoal
CERTIF - Associação para a Certificação
Kiwa Sativa – Unipessoal, Lda
CERTIS - Controlo e Certificação, Lda.
INNOQ - Instituto Nacional de Normalização e Qualidade
SGS Portugal - Sociedade Geral de Superintendência, SA
NATURALFA - Controlo e Certificação Lda. - Organismo de Certificação

In 2024, **226 IT companies** had their QMS certified according to ISO 9001 (click to find out which ones)

certificação acreditada IPAC

Software product quality

Product Quality

After all, what clients value most is **product quality** …





QUALITY
rental car

Top Quality Travel

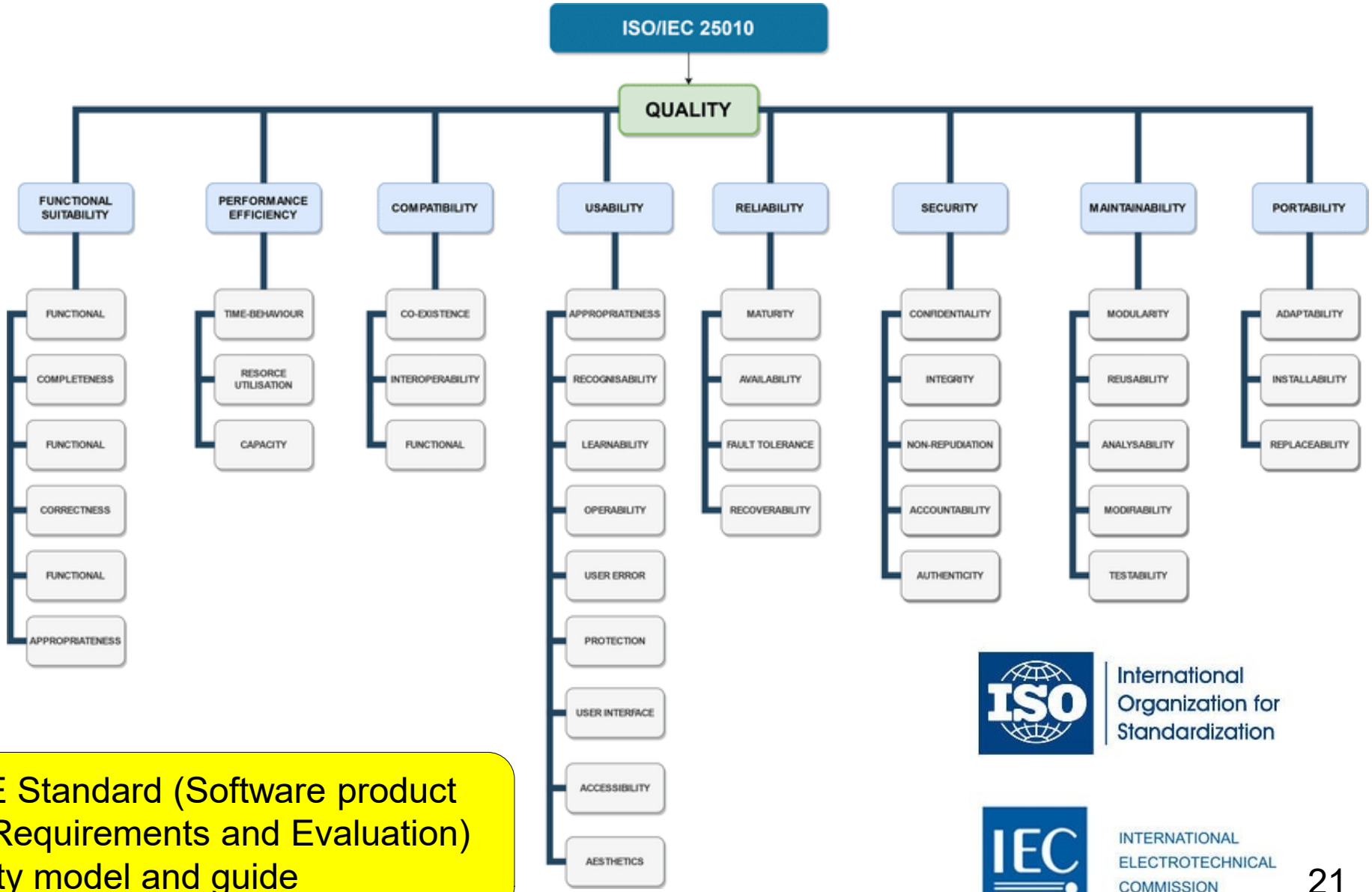








SQuaRE Standard Quality Model



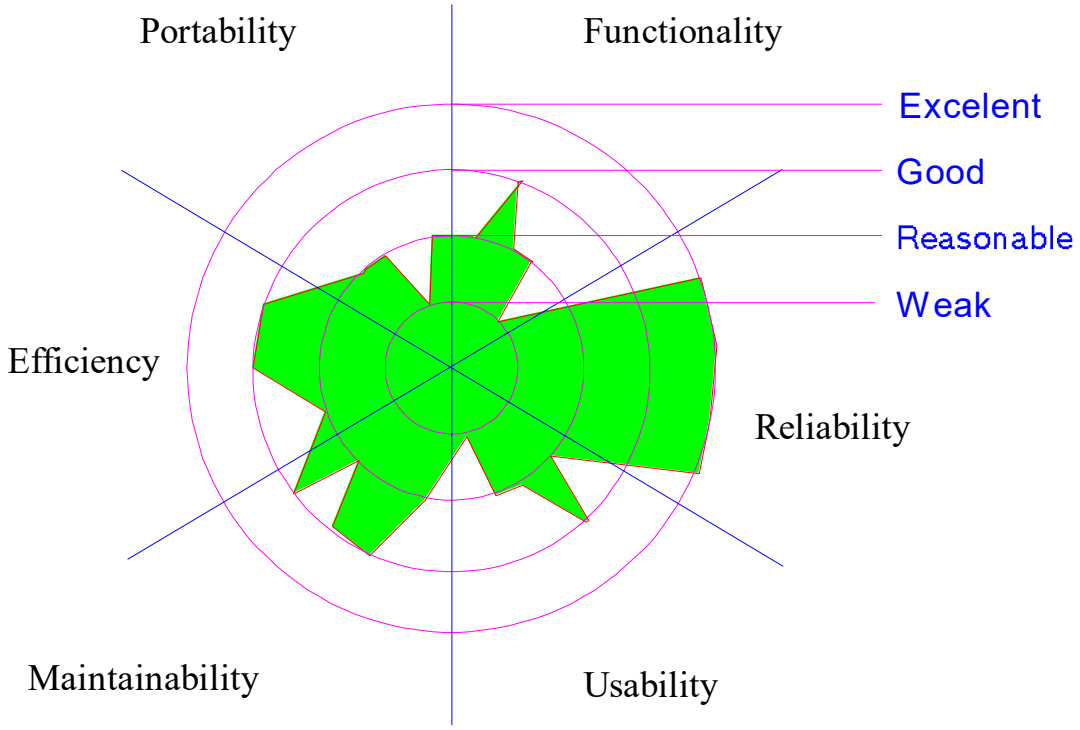
SQuaRE Standard (Software product Quality Requirements and Evaluation) — Quality model and guide

ISO International Organization for Standardization

IEC INTERNATIONAL ELECTROTECHNICAL COMMISSION

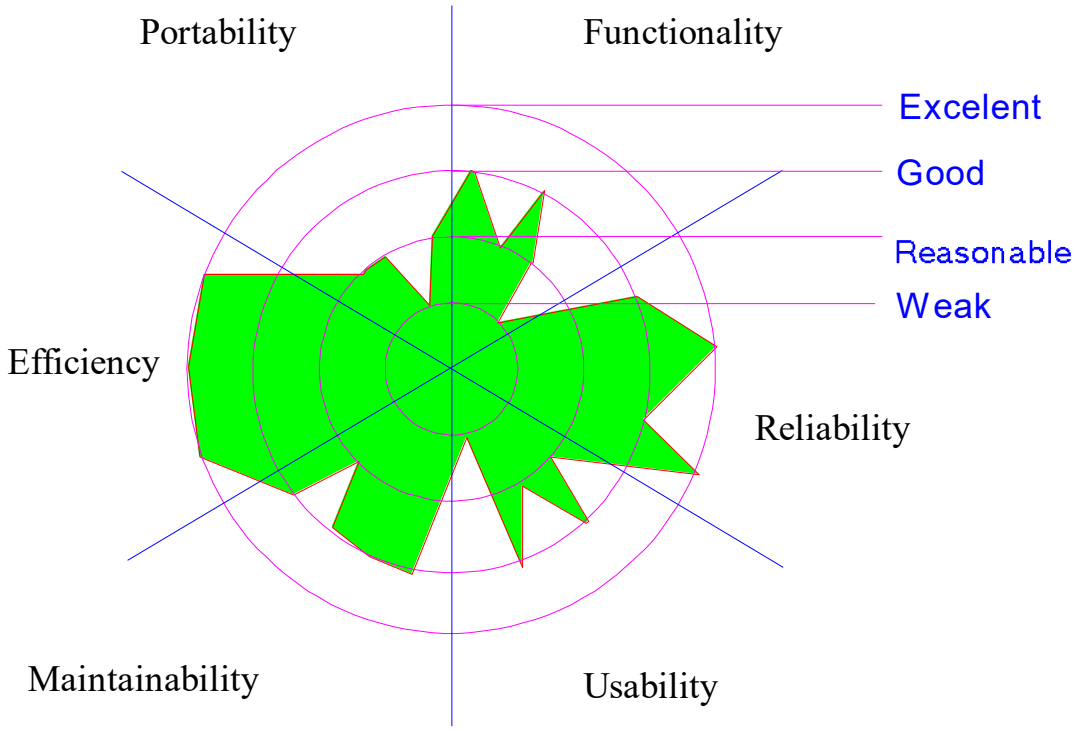
Relative importance of product quality characteristics

- Software controlling critical systems:
  - Nuclear power plant
  - Pace-maker
  - Air-traffic control
  - Railway switching



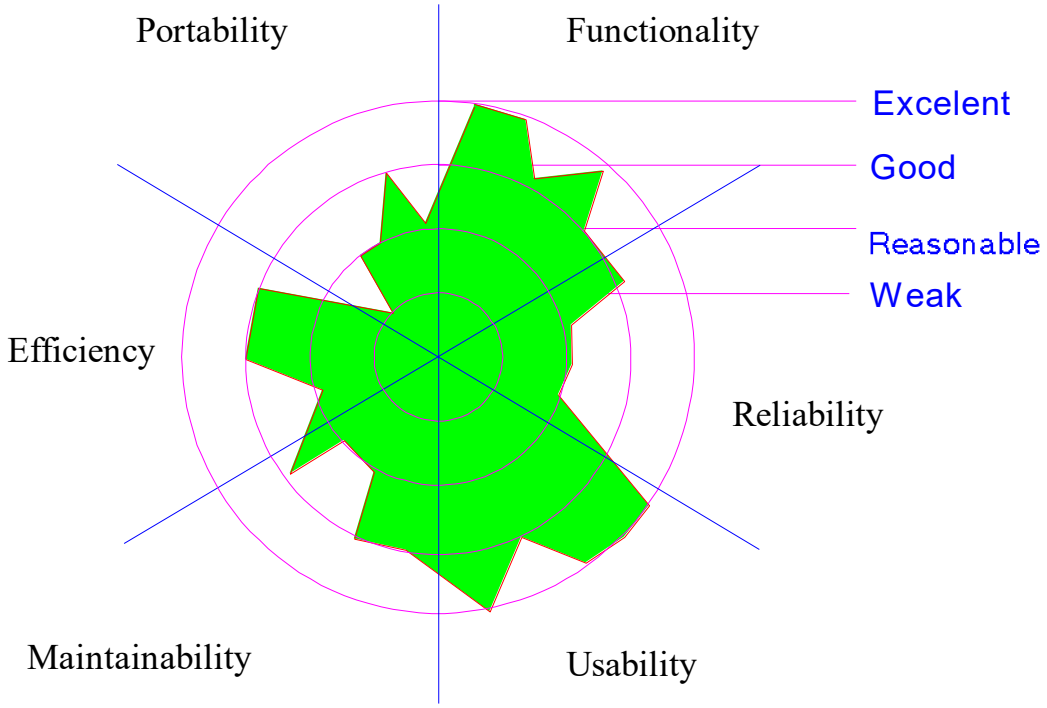
Relative importance of product quality characteristics

- Software controlling real-time systems:
  - Telecom switching
  - Communic. protocols
  - Operating systems
  - DBMS



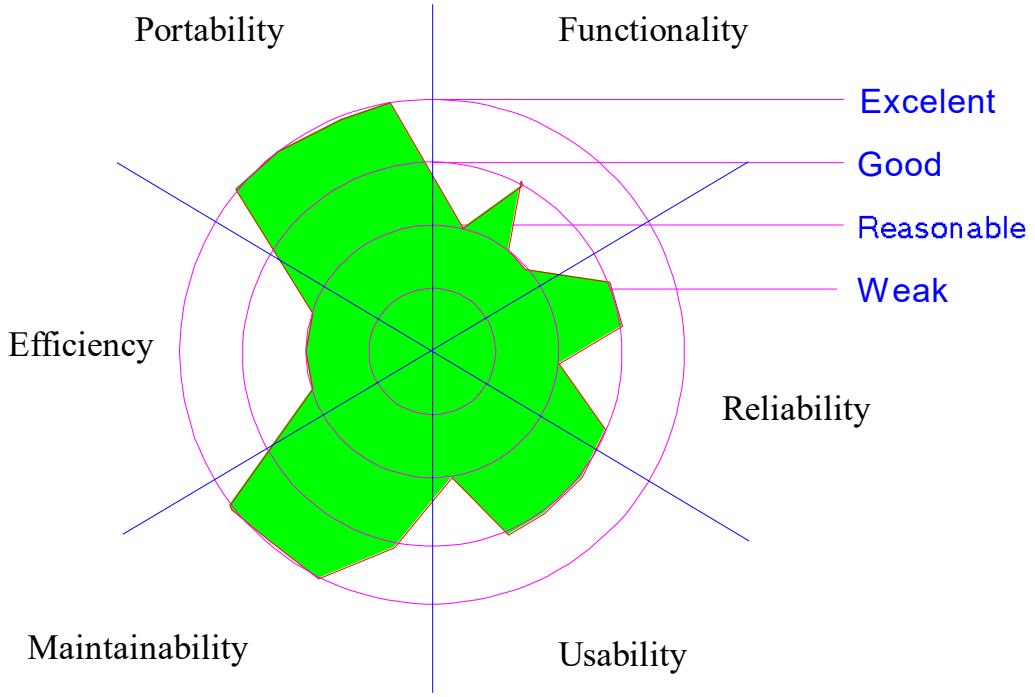
Relative importance of product quality characteristics

- Front-end systems:
  - Word processor
  - Accounts current
  - Tourist information app
  - Entertainment package



Relative importance of product quality characteristics

- Long lifecycle system:
  - Banking system
  - Air reservation system
  - Assurances system



- Project manager must consider a weighted blend!

Are product characteristics orthogonal?

	Functionality	Reliability	Usability	Efficiency	Maintain ability	Portability
Functionality			↑↓		↓	
Reliability						
Usability						
Efficiency					↓	↓
Maintain ability						↑
Portability				↓		

A DIFFICULT BALANCE:
The proposed characteristics were chosen with the concern of minimum overlap …